

Medical Statistics

Lab 5: Cleveland heart disease dataset

Introduction

Welcome to lab 5 in the Medical Statistics course. This lab is a set of unguided exercises using a new dataset. There are no separate R and SPSS versions for this lab: complete the exercises in either R or SPSS using the course methods from the previous labs.

In this lab, you will combine the methods from the previous lectures and labs to address the following research questions:

- Does maximum heart rate achieved differ across chest pain types?
- Is the presence of heart disease associated with sex or fasting blood sugar?

Dataset description

The Cleveland heart disease dataset originates from the Cleveland Clinic Foundation and focuses on heart disease diagnosis. It includes data from 303 patients on the following variables:

- **age**: Age in years
- **sex**: Sex (1 = female; 2 = male)
- **cp**: Chest pain type (1 = typical angina; 2 = atypical angina; 3 = non-anginal pain; 4 = asymptomatic)
- **trestbps**: Resting blood pressure (mm Hg at hospital admission)
- **chol**: Serum cholesterol in mg/dl
- **fbs**: Fasting blood sugar > 120 mg/dl
- **restecg**: Resting electrocardiographic results (1 = normal; 2 = ST-T wave abnormality; 3 = left ventricular hypertrophy)
- **thalach**: Maximum heart rate achieved
- **exang**: Exercise-induced angina (1 = no; 2 = yes)
- **oldpeak**: ST depression induced by exercise relative to rest
- **slope**: Slope of the peak exercise ST segment (1 = upsloping; 2 = flat; 3 = downsloping)
- **ca**: Number of major vessels (0-3) colored by fluoroscopy

- **thal**: Thallium heart scan results (1 = normal; 2 = fixed defect; 3 = reversible defect)
- **target**: Diagnosis of heart disease (1 = heart disease; 2 = no heart disease)

Exercises

Exercise 1: Download and open the dataset

Download the Cleveland heart disease dataset from the Datasets menu or from the link at the end of this page. Open or import it in R or SPSS.

Before starting the analysis, inspect the dataset and check that:

- the dataset contains 303 patients;
- all variables listed in the dataset description are present;
- categorical variables are treated as categorical variables;
- continuous variables are treated as numerical variables.

Exercise 2: Create a baseline characteristics table

Create a baseline characteristics table for the dataset.

- Include all variables except the outcome variable **target**.
- Use appropriate summaries for continuous variables, such as mean and standard deviation or median and interquartile range.
- Use frequency counts and percentages for categorical variables.
- Present the table clearly, with meaningful labels and units where appropriate.

Exercise 3: Analyze maximum heart rate across chest pain types

Research question: *Does maximum heart rate achieved differ across chest pain types?*

Use **thalach** as the outcome variable and **cp** as the grouping variable.

- Visualize the distribution of maximum heart rate achieved for each chest pain type.
- Calculate the estimated mean maximum heart rate and a 95% confidence interval for each chest pain type.
- Perform a one-way ANOVA to compare maximum heart rate across chest pain types.
- Check the assumptions for the one-way ANOVA.
- If the assumptions are not sufficiently met, perform a Kruskal-Wallis test as a non-parametric alternative.
- If the overall test is statistically significant, perform suitable post-hoc comparisons with Bonferroni-adjusted p-values.
- Write a short interpretation of the results.

Exercise 4: Analyze heart disease by sex

Research question: *Is the presence of heart disease associated with sex?*

Use **target** as the outcome variable and **sex** as the explanatory variable.

- Create a contingency table for heart disease status by sex.
- Report the row percentages.
- Check the expected cell counts.
- Select and perform the appropriate test:
 - chi-square test if the expected cell counts are sufficiently large;
 - Fisher’s exact test if the expected cell counts are too small.
- Write a short interpretation of the results.

Exercise 5: Analyze heart disease by fasting blood sugar

Research question: *Is the presence of heart disease associated with fasting blood sugar?*

Use **target** as the outcome variable and **fbs** as the explanatory variable.

- Create a contingency table for heart disease status by fasting blood sugar.
- Report the row percentages.
- Check the expected cell counts.
- Select and perform the appropriate test:
 - chi-square test if the expected cell counts are sufficiently large;
 - Fisher’s exact test if the expected cell counts are too small.
- Write a short interpretation of the results.

Downloads

[Cleveland heart disease dataset \(SPSS format\)](#)